

Priyam Srivastava

Quantum Machine Learning · Ph.D. Student, University of Pittsburgh
Indian citizen; F-1 visa (USA)

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Summary

Ph.D. student developing **quantum machine learning**—variational quantum algorithms and reinforcement learning—for the design and control of quantum systems under realistic hardware constraints. Current work spans two domains: **learned quantum sensor design** for single- and multi-parameter metrology on dipolar spin platforms, and **reinforcement-learning control** of entanglement distribution in quantum networks. Earlier training in measurement-based quantum computation and ultrafast coherent spectroscopy.

Education

Ph.D. in Information Sciences (Telecom)

Univ. of Pittsburgh (DINS, School of Computing & Information)

Jan 2024 – Present

GPA: 3.846/4.0

- Advisor: Dr. Kaushik P. Seshadreesan
- Areas: variational quantum algorithms, quantum machine learning, quantum sensing and metrology, quantum networking

B.S.–M.S. Dual Degree in Physics (Integrated)

Indian Institute of Science Education and Research, Bhopal (IISER-B), India

Jul 2017 – Jun 2022

CGPA: 8.10/10.0

- M.S. Thesis: *2-Dimensional Coherent Spectroscopy with Chirped and Unchirped Pulses*
- Advisor: Dr. Rohan Singh (Ultrafast Coherent Spectroscopy Laboratory)

Publications and Preprints

*Author order as published. **P. Srivastava** denotes my name; * denotes equal contribution.*

Peer-reviewed conference proceedings

1. **P. Srivastava**, X. Jin, J. Liu, G. Dutt, T. Purdy, K. Kim, K. P. Seshadreesan. “Variational Joint Magnetometry and Gradiometry on Dipolar Spin Chains.” *Accepted, IEEE Quantum Week (QCE) 2026, QTEM track; to appear.* arXiv:2605.03906.
2. **P. Srivastava**, V. Kumar, G. Dutt, K. P. Seshadreesan. “Variational Quantum Sensing for Structured Linear Estimation Function.” *IEEE Quantum Week (QCE) 2025.* IEEE Xplore: 11250086. (Poster track; extended journal version in preparation for QST.)

Journal submissions / preprints under review

3. X. Jin*, **P. Srivastava***, R. Wang, *et al.* (incl. K. P. Seshadreesan, J. Liu). “Four Generations of Quantum Biomedical Sensors.” *Submitted to Nature Biomedical Engineering*, 2026. arXiv:2603.29944.
4. M. Chen, Y. Gan, X. Jin, *et al.* (incl. **P. Srivastava**, K. P. Seshadreesan, J. Liu). “When AI Meets Quantum Information: A Comprehensive Review.” *Submitted to Progress in Quantum Electronics (Elsevier)*, 2026. arXiv:2607.00365.

Preprints and manuscripts in preparation

5. **P. Srivastava**, V. Kumar, G. Dutt, K. P. Seshadreesan. “Variational Probe and Measurement Optimization for Structured Phase Estimation.” *In preparation for submission to Quantum Science and Technology (QST)*. arXiv:2507.22043.
6. **P. Srivastava**, A. R. Sabavat, S. Jain, *et al.* (incl. K. P. Seshadreesan). “Sequential vs. Simultaneous Entanglement Swapping under Optimal Link-Layer Control.” *In preparation for submission to ACM Conf. on Post-Quantum and Quantum-based Security (PQQS) 2026*. arXiv:2605.04047.
7. **P. Srivastava**, V. Katyal, A. Raina. “Encoder Circuit for a Surface Code using a Measurement-Based Quantum Computing Model.” Preprint, 2023. arXiv:2306.10267.

Awards, Grants, and Collaborations

- **PQI Community Collaboration Award (2026)** — “Dynamic Programming & Reinforcement Learning for Quantum Network Control (DyReQNet-Ctrl),” significant contributor. PI Dr. K. P. Seshadreesan; co-PIs Profs. P. Krishnamurthy, A. Scheller-Wolf, S. Tayur.
- **PQI Community Collaboration Award (2024)** — “NV-Center-Based Quantum Sensing for Biomedical Ultrasound Imaging,” contributor. PI Prof. Junyu Liu; co-PIs Profs. K. P. Seshadreesan, G. Dutt, K. Kang.

Research Experience

Graduate Student Researcher — Quantum Information & Networking Group Jan 2024 – Present

University of Pittsburgh Advisor: Dr. Kaushik P. Seshadreesan

- Joint magnetometry/gradiometry on dipolar spin chains using $\det(F)$ as the multiparameter objective; motivated by the rank-deficient QFI of GHZ states [1].
- RL link-layer agents + network-layer controllers for sequential vs. simultaneous entanglement swapping; identified a coherence-time regime structure in T_c^{ext}/τ governing the connection-less penalty [6].
- Variational quantum sensing for structured phase estimation (CMA-ES on dipolar-gate ansätze); learned probes approach entanglement-enhanced precision bounds [5, 2].
- Co-developed a generational framework (G1–G4) for quantum biomedical sensors [3].

Research Project — MBQC and Surface Codes

Jun 2022 – Jun 2023

IISER Bhopal Advisor: Dr. Ankur Raina

- Encoder circuit for a surface code via the MBQC model; Qiskit simulations on AWS Braket [7].

M.S. Thesis — Ultrafast Coherent Spectroscopy

Aug 2021 – May 2022

IISER Bhopal Advisor: Dr. Rohan Singh

- Numerical (Python) study of linear chirp in two- and three-level systems via the optical Bloch equations.
- Theory of 2D coherent spectroscopy; bench-side experience in ultrafast spectroscopy setup.

Summer Research Internship — Entanglement Entropy

May 2019 – Jul 2019

IIT Allahabad Advisor: Dr. Srijit Bhattacharjee

- Entanglement entropy in coupled harmonic oscillator systems.

Talks and Presentations

- **Contributed talk**, “Variational Probe and Measurement Optimization for Structured Phase Estimation,” Workshop on Frontiers in Quantum Information & Technologies (FQIT 2026), Indian Institute of Information Technology Allahabad, March 26–29, 2026.

- **Poster**, “Variational Quantum Sensing for Structured Linear Estimation Function,” IEEE International Conference on Quantum Computing and Engineering (QCE 2025).

Teaching

- **Recitation Instructor & Grader TA**, CS 1613 *Modern Quantum Computing*, University of Pittsburgh, Spring 2026.
- **Lab Teaching Assistant**, CS 0011 *Introduction to Computing for Scientists*, University of Pittsburgh, Spring 2026.
- **Teaching Assistant**, CS 1900 *Internship* and CS 1980 *Team Project*, University of Pittsburgh, Fall 2025 – Spring 2026.

Service and Professional Development

- **Reviewer**, *npj Quantum Information* (Nature Portfolio), 2025.
- **Student Volunteer**, IEEE International Conference on Quantum Computing and Engineering (QCE), 2025.
- **QNumerics 2026** — Numerical Methods in Quantum Information Science, UMass Amherst, June 2026.
- **STAQ Summer School on Quantum Information Theory and Computation**, Duke University, June 2024.

Certifications

- **IBM Quantum Learning** — Variational Algorithm Design & Basics of Quantum Information (badges), 2024–2025.

Technical Skills

Programming: Python (NumPy, SciPy, JAX, PyTorch, Gymnasium), C, MATLAB, Mathematica; HPC workflows on Pitt CRC (SLURM array jobs).

Quantum SDKs: PennyLane, Qiskit, AWS Braket.

Methods: Variational quantum algorithms, parameterized quantum circuits, CMA-ES and gradient-free optimization, quantum and classical Fisher information, multiparameter estimation, reinforcement learning, stabilizer formalism, MBQC.

Hardware contexts: NV centers in diamond, ultrafast/nonlinear optics.

Languages: English (fluent), Hindi (native).